

BIOINFORMATICS ASSIGNMENT

P6

Name: Sidharthan SC

Roll No: BE21B039

1. The given IDs in Set were mapped using UniProt and the sequences were downloaded in FASTA format.

Retrieve/ID mapping

Enter one of more IDs (100,000 max). You may also load from a text file. Separate IDs by whitespace (space, tab, newline) or commas.

TPIS_HUMAN, TPIS_YEAST, TPIS_GRAGA, TPIS_TRYCR, TPIS_MAIZE, TPIS_MOUSE, TPIS_DROME, TPIS_RABIT, TPIS_CAEEL

Your input contains 9 IDs

From database: UniProtKB AC-ID

To database: UniProtKB

Name your ID Mapping job

TPIS_HUMAN +8 UniProtKB_AC-ID -> UniProtKB

Reset Map 9 IDs

ID mapping 9 results found for UniProtKB_AC-ID -> UniProtKB

Overview Input Parameters API Request

9 IDs were mapped to 9 results

From	Entry	Entry Name	Protein Names	Gene Names	Organism	Length
TPIS_HUMAN	P69174	TPIS_HUMAN	Triosephosphate isomerase[...]	TPH1, TPI	Homo sapiens (Human)	249 AA
TPIS_YEAST	P00942	TPIS_YEAST	Triosephosphate isomerase[...]	TPH1, YDR050C, Y09609.05C	Saccharomyces cerevisiae (strain ATCC 204508 / S288c) (Baker's yeast)	248 AA
TPIS_GRAGA	P48492	TPIS_GRAGA	Triosephosphate isomerase, cytosolic[...]	TPH1	Gracilaria gracilis (Red alga)	259 AA
TPIS_TRYCR	P52270	TPIS_TRYCR	Triosephosphate isomerase, glycosomal[...]		Trypanosoma cruzi	251 AA
TPIS_MAIZE	P12863	TPIS_MAIZE	Triosephosphate isomerase, cytosolic[...]		Zea mays (Maize)	253 AA
TPIS_MOUSE	P17751	TPIS_MOUSE	Triosephosphate isomerase[...]	Tpi1, Tpi	Mus musculus (Mouse)	249 AA
TPIS_DROME	P29613	TPIS_DROME	Triosephosphate isomerase[...]	Tpi, CG2171	Drosophila melanogaster (Fruit fly)	247 AA
TPIS_RABIT	P00939	TPIS_RABIT	Triosephosphate isomerase[...]	TPH1	Oryctolagus cuniculus (Rabbit)	249 AA
TPIS_CAEEL	Q10657	TPIS_CAEEL	Triosephosphate isomerase[...]	tpi-1, Y1707B.7	Caenorhabditis elegans	247 AA

Using fasta file we align it using clustal omega

SET 1:

POSITION	ALIGNMENT	SCORE 1	SCORE 2	SCORE 3	SCORE 4	SCORE 5
1	:	0	1.021	5	1.019	0.943
2	S	-0.305	0.865	3.008	0.729	0.253
3	L	0	0.909	4	0.916	0.943
4	S	0	0.953	4	0.951	0.943
5	D	-1.121	0.527	1.149	0.492	-1.597
6	K	-1.16	0.561	1.157	0.439	-1.685
7	D	0	0.973	6	0.972	0.943
8	K	0	0.956	5	0.95	0.943
9	A	-1.169	0.541	1.868	0.465	-1.705
10	A	-0.305	0.917	4.661	0.77	0.253
11	V	-0.586	0.744	3.603	0.763	-0.385
12	R	-0.305	0.871	4.504	0.744	0.253
13	A	-1.034	0.567	1.909	0.566	-1.399
14	L	-1.16	0.559	1.521	0.415	-1.685
15	W	-0.305	0.943	9.306	0.888	0.253
16	S	-0.916	0.612	2.264	0.604	-1.134
17	K	0	0.956	5	0.95	0.943
18	I	-0.689	0.691	3.504	0.719	-0.618
19	G	-0.305	0.886	4.992	0.831	0.253
20	K	-1.034	0.613	2.405	0.502	-1.399
21	S	-0.886	0.713	4.372	0.627	-1.064
22	A	-0.474	0.74	2.876	0.767	-0.131
23	D	-1.034	0.624	2.24	0.517	-1.399
24	A	-0.908	0.665	2.455	0.593	-1.114
25	I	-0.886	0.757	4.314	0.679	-1.064
26	G	0	0.977	6	0.977	0.943
27	N	-0.886	0.65	1.826	0.561	-1.064
28	D	-0.305	0.922	4.512	0.784	0.253
29	A	-0.305	0.817	3.347	0.768	0.253
30	L	0	0.909	4	0.916	0.943
31	S	-0.886	0.749	3.298	0.628	-1.064
32	R	0	1.015	5	1.011	0.943

33	M	-0.305	0.933	4.008	0.944	0.253
34	I	-0.305	0.904	4.992	0.78	0.253
35	V	-1.295	0.47	0.959	0.373	-1.991
36	V	-1.367	0.472	1.132	0.386	-2.154
37	Y	-0.474	0.83	5.14	0.718	-0.131
38	P	0	0.993	7	0.993	0.943
39	Q	-0.474	0.821	3.215	0.715	-0.131
40	T	0	0.977	5	0.979	0.943
41	K	0	0.956	5	0.95	0.943
42	T	0	0.977	5	0.979	0.943
43	Y	0	1.016	7	1.012	0.943
44	F	0	0.988	6	0.989	0.943
45	S	-0.474	0.822	4.521	0.712	-0.131
46	H	0	0.969	8	0.972	0.943
47	W	-0.6	0.823	4.24	0.711	-0.417
48	P	-0.857	0.657	2.295	0.582	-1
49	D	-0.305	0.888	5.174	0.901	0.253
50	V	-0.586	0.674	2.81	0.621	-0.385
51	T	-0.6	0.776	2.752	0.605	-0.417
52	P	-0.586	0.745	3.959	0.689	-0.385
53	G	0	0.977	6	0.977	0.943
54	S	0	0.953	4	0.951	0.943
55	P	-0.6	0.73	2.322	0.628	-0.417
56	H	-0.305	0.936	4.198	0.8	0.253
57	I	-0.305	0.872	3.835	0.748	0.253
58	K	0	0.956	5	0.95	0.943
59	A	-0.86	0.596	1.835	0.687	-1.006
60	H	0	0.969	8	0.972	0.943
61	G	0	0.977	6	0.977	0.943
62	K	-0.6	0.777	3.405	0.812	-0.417
63	K	0	0.956	5	0.95	0.943
64	V	0	0.956	4	0.956	0.943
65	M	-0.76	0.66	2.124	0.612	-0.778
66	G	-0.76	0.704	2.694	0.561	-0.778
67	G	-0.305	0.817	3.355	0.692	0.253

68	I	-0.305	0.825	3.669	0.71	0.253
69	A	-0.86	0.643	2.19	0.545	-1.006
70	L	-1.673	0.36	0.124	0.382	-2.848
71	A	0	0.906	4	0.909	0.943
72	V	-0.586	0.701	2.413	0.682	-0.385
73	S	-1.594	0.406	0.934	0.359	-2.67
74	K	-0.305	0.879	6.488	0.745	0.253
75	I	-0.995	0.543	2.612	0.534	-1.312
76	D	-0.305	0.888	5.331	0.888	0.253
77	D	0	0.973	6	0.972	0.943
78	L	-0.76	0.675	3.14	0.683	-0.778
79	K	-0.6	0.807	4.446	0.618	-0.417
80	T	-0.76	0.733	3.149	0.664	-0.778
81	G	-0.305	0.817	3.355	0.692	0.253
82	L	0	0.909	4	0.916	0.943
83	M	-0.305	0.869	3.182	0.749	0.253
84	E	-1.414	0.451	0.678	0.428	-2.262
85	L	0	0.909	4	0.916	0.943
86	S	0	0.953	4	0.951	0.943
87	E	-0.305	0.888	5.331	0.763	0.253
88	Q	-0.305	0.826	3.017	0.715	0.253
89	H	0	0.969	8	0.972	0.943
90	A	0	0.906	4	0.909	0.943
91	Y	-0.76	0.731	4.967	0.618	-0.778
92	K	0	0.956	5	0.95	0.943
93	L	0	0.909	4	0.916	0.943
94	R	0	1.015	5	1.011	0.943
95	V	0	0.956	4	0.956	0.943
96	D	0	0.973	6	0.972	0.943
97	P	0	0.993	7	0.993	0.943
98	A	-0.305	0.86	3.339	0.715	0.253
99	N	0	1.012	6	1.011	0.943
100	F	0	0.988	6	0.989	0.943
101	K	0	0.956	5	0.95	0.943
102	I	-0.6	0.739	3.058	0.632	-0.417

103	L	0	0.909	4	0.916	0.943
104	N	-0.6	0.781	2.926	0.619	-0.417
105	H	-0.305	0.886	6.653	0.839	0.253
106	C	-0.305	0.937	7.306	0.962	0.253
107	I	-0.6	0.739	3.058	0.588	-0.417
108	L	-0.305	0.825	3.678	0.783	0.253
109	V	-0.474	0.784	2.215	0.844	-0.131
110	V	-0.474	0.806	3.479	0.677	-0.131
111	I	-0.6	0.735	3.24	0.58	-0.417
112	S	-0.305	0.814	3.504	0.683	0.253
113	T	-1.72	0.345	0.702	0.355	-2.955
114	M	-0.305	0.885	6.322	0.769	0.253
115	F	-0.935	0.559	1.207	0.529	-1.175
116	P	0	0.993	7	0.993	0.943
117	K	-1.846	0.302	0.562	0.299	-3.241
118	E	-0.916	0.633	3.008	0.584	-1.134
119	F	-0.305	0.893	4.992	0.837	0.253
120	T	-0.305	0.891	4.182	0.907	0.253
121	P	0	0.993	7	0.993	0.943
122	E	-0.76	0.66	2.182	0.589	-0.778
123	A	-0.886	0.695	2.736	0.519	-1.064
124	H	0	0.969	8	0.972	0.943
125	V	-0.305	0.814	3.339	0.685	0.253
126	S	0	0.953	4	0.951	0.943
127	L	0	0.909	4	0.916	0.943
128	D	0	0.973	6	0.972	0.943
129	K	0	0.956	5	0.95	0.943
130	F	0	0.988	6	0.989	0.943
131	L	-0.305	0.822	3.355	0.839	0.253
132	S	-0.86	0.595	2.289	0.528	-1.006
133	G	-1.414	0.459	1.496	0.4	-2.262
134	V	0	0.956	4	0.956	0.943
135	A	-0.6	0.769	2.909	0.577	-0.417
136	L	-0.6	0.796	3.231	0.615	-0.417
137	A	-0.305	0.86	3.339	0.715	0.253

138	L	0	0.909	4	0.916	0.943
139	A	-0.305	0.881	4.165	0.738	0.253
140	E	-0.305	0.868	3.347	0.743	0.253
141	R	-0.305	0.871	4.504	0.744	0.253
142	Y	0	1.016	7	1.012	0.943
143	R	0	1.015	5	1.011	0.943

The same steps were followed for the set 2 also and following table is the result

SET 2:

POSITION	ALIGNMENT	SCORE 1	SCORE 2	SCORE 3	SCORE 4	SCORE 5
1	M	-1.216 *	0.529 *	1.306 *	0.494 *	-1.000 *
2	A	-1.216 *	0.529 *	1.306 *	0.494 *	-1.000 *
3	S	-0.974	0.648	1.328	0.666	-0.56
4	K	-1.494	0.426	1.109	0.365	-1.507
5	P	-0.377	0.882	3.5	0.827	0.529
6	Q	-0.736	0.726	2.938	0.641	-0.125
7	P	-0.377	0.884	3.828	0.83	0.529
8	I	-0.736	0.769	3.172	0.713	-0.125
9	A	-0.377	0.801	3.125	0.739	0.529
10	A	-0.377	0.812	4.656	0.754	0.529
11	A	-0.377	0.812	4.656	0.754	0.529
12	N	0	0.995	6	0.994	1.215
13	W	-0.349	0.909	8.963	0.876	0.58
14	K	0	0.962	5	0.966	1.215
15	C	-0.937	0.679	2.395	0.639	-0.492
16	N	0	0.995	6	0.994	1.215
17	G	-0.349	0.835	4	0.81	0.58
18	S	-1.311	0.51	0.79	0.53	-1.173
19	E	-1.303	0.553	1.568	0.515	-1.159
20	S	-1.523	0.401	0.901	0.372	-1.56
21	L	-1.581	0.481	0.519	0.482	-1.666
22	L	-1.061	0.526	2.667	0.514	-0.718
23	V	-1.677	0.317	0.198	0.27	-1.841
24	P	-1.003	0.64	2.025	0.597	-0.612
25	L	-0.687	0.664	3.012	0.656	-0.036
26	I	-0.965	0.55	2.716	0.557	-0.543

27	E	-1.735	0.362	0.136	0.393	-1.946
28	T	-1.003	0.663	1.815	0.595	-0.612
29	L	-0.349	0.859	3.235	0.826	0.58
30	N	-0.349	0.888	4.988	0.885	0.58
31	A	-1.003	0.591	1.951	0.549	-0.612
32	A	-0.684	0.691	2.716	0.676	-0.031
33	T	-1.889	0.305	0.383	0.302	-2.227
34	F	-1.581	0.426	0.457	0.364	-1.666
35	D	-1.149	0.604	1.543	0.553	-0.879
36	H	-1.677	0.372	0.383	0.348	-1.841
37	D	-1.523	0.429	1.259	0.421	-1.56
38	:	-1.216 *	0.529 *	1.306 *	0.494 *	-1.000 *
39	:	-1.216 *	0.529 *	1.306 *	0.494 *	-1.000 *
40	V	-0.687	0.645	2.222	0.684	-0.036
41	Q	-0.849	0.671	3.432	0.623	-0.331
42	C	-0.349	0.829	3.074	0.795	0.58
43	V	0	0.93	4	0.93	1.215
44	V	-1.061	0.545	2.136	0.563	-0.718
45	A	-1.003	0.594	2.062	0.557	-0.612
46	P	-0.349	0.893	5.049	0.894	0.58
47	T	-0.349	0.889	5.395	0.855	0.58
48	F	-1.215	0.498	0.889	0.51	-0.998
49	L	-1.677	0.321	0.593	0.294	-1.841
50	H	-0.684	0.808	5.247	0.771	-0.031
51	I	-0.687	0.664	3.012	0.686	-0.036
52	P	-1.149	0.595	1.272	0.564	-0.879
53	M	-1.465	0.504	1.704	0.492	-1.454
54	T	-1.149	0.509	1.704	0.45	-0.879
55	K	-0.937	0.625	2.37	0.582	-0.492
56	A	-1.215	0.541	1.556	0.505	-0.998
57	R	-1.427	0.462	0.889	0.399	-1.385
58	L	-0.349	0.85	3.407	0.809	0.58
59	T	-1.216 *	0.529 *	1.306 *	0.494 *	-1.000 *
60	N	-1.523	0.455	0.877	0.44	-1.56
61	P	-1.149	0.584	1.667	0.551	-0.879

62	K	-1.427	0.467	1.469	0.421	-1.385
63	F	-1.311	0.477	1.852	0.497	-1.173
64	Q	-1.831	0.321	-0.222	0.292	-2.121
65	I	-1.003	0.611	2.531	0.549	-0.612
66	A	-0.684	0.691	2.716	0.623	-0.031
67	A	-0.349	0.798	3.235	0.793	0.58
68	Q	0	0.994	5	0.993	1.215
69	N	0	0.995	6	0.994	1.215
70	A	-0.687	0.669	3.568	0.649	-0.036
71	I	-0.849	0.723	4	0.666	-0.331
72	:	-1.074	0.596	1.234	0.506	-0.741
73	T	-1.149	0.532	0.827	0.465	-0.879
74	R	-1.523	0.408	0.654	0.38	-1.56
75	S	-1.369	0.453	1.358	0.446	-1.279
76	G	0	0.944	6	0.951	1.215
77	A	0	0.909	4	0.905	1.215
78	F	0	1.001	6	1.006	1.215
79	T	0	0.983	5	0.984	1.215
80	G	0	0.944	6	0.951	1.215
81	E	0	0.962	5	0.96	1.215
82	V	-1.149	0.559	2	0.478	-0.879
83	S	-0.349	0.873	3.235	0.84	0.58
84	L	-1.149	0.562	1.568	0.469	-0.879
85	Q	-1.465	0.391	0.741	0.375	-1.454
86	I	-0.684	0.811	3.235	0.744	-0.031
87	L	-0.937	0.601	2.914	0.557	-0.492
88	K	-0.349	0.849	3.605	0.848	0.58
89	D	-0.349	0.883	5.012	0.878	0.58
90	Y	-1.523	0.43	1.123	0.415	-1.56
91	G	0	0.944	6	0.951	1.215
92	I	-1.003	0.58	1.778	0.529	-0.612
93	S	-1.677	0.406	0.778	0.368	-1.841
94	W	0	1.013	11	1.013	1.215
95	V	0	0.93	4	0.93	1.215
96	V	-0.687	0.644	3.506	0.676	-0.036

97	L	0	0.963	4	0.959	1.215
98	G	0	0.944	6	0.951	1.215
99	H	0	1.017	8	1.017	1.215
100	S	0	0.979	4	0.977	1.215
101	E	0	0.962	5	0.96	1.215
102	R	0	0.999	5	1	1.215
103	R	0	0.999	5	1	1.215
104	L	-1.149	0.592	1.728	0.522	-0.879
105	Y	-1.273	0.484	1.827	0.463	-1.104
106	:	-1.216 *	0.529 *	1.306 *	0.494 *	-1.000 *
107	:	-1.216 *	0.529 *	1.306 *	0.494 *	-1.000 *
108	Y	-0.849	0.703	3.346	0.65	-0.331
109	G	-0.684	0.735	3.074	0.682	-0.031
110	E	0	0.962	5	0.96	1.215
111	T	-0.684	0.768	2.704	0.7	-0.031
112	N	-0.53	0.796	4.272	0.765	0.25
113	E	-1.149	0.546	1.605	0.484	-0.879
114	I	-1.149	0.577	1.642	0.508	-0.879
115	V	-0.53	0.762	3.654	0.726	0.25
116	A	-0.687	0.601	2.42	0.643	-0.036
117	E	-1.311	0.505	1.914	0.499	-1.173
118	K	0	0.962	5	0.966	1.215
119	V	-0.995	0.535	1.679	0.492	-0.598
120	A	-1.303	0.475	1.21	0.465	-1.159
121	Q	-1.149	0.638	3.346	0.569	-0.879
122	A	0	0.909	4	0.905	1.215
123	C	-0.349	0.861	3.074	0.824	0.58
124	A	-1.427	0.419	1.136	0.381	-1.385
125	:	-1.216 *	0.529 *	1.306 *	0.494 *	-1.000 *
126	A	-1.061	0.514	1.63	0.491	-0.718
127	G	-0.349	0.837	4.79	0.813	0.58
128	F	-1.003	0.642	2.444	0.566	-0.612
129	H	-0.937	0.592	1.444	0.573	-0.492
130	V	0	0.93	4	0.93	1.215
131	I	-0.349	0.858	3.802	0.849	0.58

132	V	-1.303	0.492	0.901	0.401	-1.159
133	C	0	1.017	9	1.018	1.215
134	V	-0.53	0.762	3.654	0.726	0.25
135	G	0	0.944	6	0.951	1.215
136	E	0	0.962	5	0.96	1.215
137	T	-0.965	0.592	1.333	0.598	-0.543
138	N	-0.349	0.859	2.642	0.819	0.58
139	E	-0.637	0.706	3.778	0.782	0.056
140	E	-0.349	0.857	4.407	0.85	0.58
141	R	-0.349	0.89	4.407	0.856	0.58
142	E	-0.349	0.853	4.21	0.816	0.58
143	A	-0.349	0.802	3.407	0.766	0.58
144	G	0	0.944	6	0.951	1.215
145	R	-1.677	0.414	-0.222	0.387	-1.841
146	T	0	0.983	5	0.984	1.215
147	A	-1.831	0.35	-0.062	0.283	-2.121
148	A	-1.311	0.452	0.988	0.454	-1.173
149	V	0	0.93	4	0.93	1.215
150	V	-0.684	0.726	1.84	0.692	-0.031
151	L	-1.273	0.497	0.395	0.44	-1.104
152	T	-1.215	0.533	1.309	0.562	-0.998
153	Q	0	0.994	5	0.993	1.215
154	L	-0.965	0.604	1.543	0.623	-0.543
155	A	-1.427	0.481	0.951	0.424	-1.385
156	A	-0.637	0.629	2.222	0.712	0.056
157	V	-1.149	0.562	2.407	0.491	-0.879
158	A	-1.003	0.581	1.852	0.507	-0.612
159	Q	-0.995	0.612	3.037	0.569	-0.598
160	K	-1.215	0.519	1.593	0.511	-0.998
161	L	-1.215	0.484	1.802	0.458	-0.998
162	S	-1.216 *	0.529 *	1.306 *	0.494 *	-1.000 *
163	K	-1.216 *	0.529 *	1.306 *	0.494 *	-1.000 *
164	E	-1.003	0.635	2.136	0.568	-0.612
165	A	-0.684	0.766	3.407	0.718	-0.031
166	W	0	1.013	11	1.013	1.215

167	S	-1.003	0.658	2.086	0.617	-0.612
168	R	-1.215	0.549	2	0.558	-0.998
169	V	-0.349	0.822	3.802	0.815	0.58
170	V	0	0.93	4	0.93	1.215
171	I	-1.099	0.495	2.667	0.518	-0.787
172	A	0	0.909	4	0.905	1.215
173	Y	0	1.013	7	1.011	1.215
174	E	0	0.962	5	0.96	1.215
175	P	0	0.995	7	0.997	1.215
176	V	0	0.93	4	0.93	1.215
177	W	0	1.013	11	1.013	1.215
178	A	0	0.909	4	0.905	1.215
179	I	0	0.971	4	0.97	1.215
180	G	0	0.944	6	0.951	1.215
181	T	0	0.983	5	0.984	1.215
182	G	0	0.944	6	0.951	1.215
183	K	-0.684	0.75	2.914	0.718	-0.031
184	V	-0.937	0.591	2.037	0.535	-0.492
185	A	0	0.909	4	0.905	1.215
186	T	-0.349	0.876	4.198	0.87	0.58
187	P	-0.349	0.884	5.21	0.88	0.58
188	Q	-1.215	0.53	2.185	0.514	-0.998
189	Q	-0.349	0.888	4.025	0.853	0.58
190	A	0	0.909	4	0.905	1.215
191	Q	-0.349	0.885	4.407	0.852	0.58
192	E	-0.684	0.753	3.852	0.689	-0.031
193	V	-0.349	0.822	3.802	0.788	0.58
194	H	0	1.017	8	1.017	1.215
195	E	-0.637	0.669	2.222	0.641	0.056
196	L	-1.677	0.389	-0.074	0.351	-1.841
197	L	-0.849	0.656	2.914	0.592	-0.331
198	R	0	0.999	5	1	1.215
199	R	-1.677	0.35	0.383	0.319	-1.841
200	W	-0.53	0.822	7.296	0.792	0.25
201	V	-0.349	0.85	3.407	0.808	0.58

202	R	-1.149	0.545	1.827	0.476	-0.879
203	S	-1.303	0.564	1.778	0.525	-1.159
204	K	-0.637	0.724	3.222	0.69	0.056
205	L	-1.149	0.516	2.173	0.471	-0.879
206	G	-0.849	0.673	2.296	0.656	-0.331
207	T	-1.215	0.555	1.42	0.532	-0.998
208	D	-1.677	0.333	0.543	0.319	-1.841
209	I	-0.684	0.707	3.012	0.633	-0.031
210	A	-0.349	0.802	3.407	0.796	0.58
211	A	-1.465	0.426	1.012	0.405	-1.454
212	Q	-1.003	0.651	2.099	0.575	-0.612
213	L	-0.637	0.713	2.222	0.68	0.056
214	R	0	0.999	5	1	1.215
215	I	0	0.971	4	0.97	1.215
216	L	-0.937	0.612	1.963	0.586	-0.492
217	Y	0	1.013	7	1.011	1.215
218	G	0	0.944	6	0.951	1.215
219	G	0	0.944	6	0.951	1.215
220	S	0	0.979	4	0.977	1.215
221	V	-0.349	0.815	3.21	0.776	0.58
222	T	-0.684	0.773	3.346	0.711	-0.031
223	A	-0.965	0.53	1.765	0.521	-0.543
224	K	-1.003	0.592	1.642	0.518	-0.612
225	N	-0.637	0.733	3.222	0.814	0.056
226	A	-0.687	0.669	3.568	0.65	-0.036
227	R	-1.303	0.523	1.593	0.429	-1.159
228	T	-0.849	0.666	2.272	0.596	-0.331
229	L	-0.349	0.859	3.235	0.823	0.58
230	Y	-1.003	0.59	1.383	0.52	-0.612
231	Q	-1.465	0.444	1.111	0.427	-1.454
232	M	-0.937	0.637	2.457	0.601	-0.492
233	R	-0.684	0.777	3.802	0.707	-0.031
234	D	-0.349	0.883	5.012	0.852	0.58
235	I	-0.687	0.637	3.506	0.638	-0.036
236	N	-0.349	0.883	5.012	0.847	0.58

237	G	0	0.944	6	0.951	1.215
238	F	0	1.001	6	1.006	1.215
239	L	0	0.963	4	0.959	1.215
240	V	0	0.93	4	0.93	1.215
241	G	0	0.944	6	0.951	1.215
242	G	0	0.944	6	0.951	1.215
243	A	0	0.909	4	0.905	1.215
244	S	0	0.979	4	0.977	1.215
245	L	-0.349	0.862	3.617	0.829	0.58
246	K	0	0.962	5	0.966	1.215
247	P	0	0.995	7	0.997	1.215
248	E	-0.684	0.752	3.309	0.713	-0.031
249	F	0	1.001	6	1.006	1.215
250	V	-0.684	0.715	3.259	0.678	-0.031
251	E	-0.849	0.682	3.469	0.628	-0.331
252	I	0	0.971	4	0.97	1.215
253	I	-0.349	0.858	3.802	0.824	0.58
254	E	-0.684	0.782	3.988	0.714	-0.031
255	A	-0.53	0.714	2.963	0.663	0.25
256	T	-1.465	0.462	1.062	0.457	-1.454
257	K	-1.427	0.502	1.519	0.451	-1.385
258	:	-1.216 *	0.529 *	1.306 *	0.494 *	-1.000 *
259	:	-1.216 *	0.529 *	1.306 *	0.494 *	-1.000 *
260	:	-1.216 *	0.529 *	1.306 *	0.494 *	-1.000 *
261	:	-1.216 *	0.529 *	1.306 *	0.494 *	-1.000 *
262	:	-1.216 *	0.529 *	1.306 *	0.494 *	-1.000 *
263	:	-1.216 *	0.529 *	1.306 *	0.494 *	-1.000 *
264	:	-1.216 *	0.529 *	1.306 *	0.494 *	-1.000 *

* gap fraction no less than 0.50; conservation set to M-S
M: mean; S: standard deviation

2.

The highest-scoring top 10 residues as follows

SET 1:

POSITION	ALIGNMENT	SCORE
1	:	0
3	L	0
4	S	0
7	D	0
8	K	0
17	K	0
26	G	0
30	L	0
32	R	0
38	P	0

SET 2:

POSITION	ALIGNMENT	SCORE
12	N	0
14	K	0
16	N	0
43	V	0
68	Q	0
69	N	0
76	G	0
77	A	0
78	F	0
79	T	0

The lowest-scoring top 10 residues as follows

SET 1:

POSITION	ALIGNMENT	SCORE
117	K	-1.846
113	T	-1.72
70	L	-1.673
73	S	-1.594
84	E	-1.414
133	G	-1.414
36	V	-1.367
35	V	-1.295
9	A	-1.169
6	K	-1.16

SET 2:

POSITION	ALIGNMENT	SCORE
33	T	-1.889
64	Q	-1.831
147	A	-1.831
27	E	-1.735
23	V	-1.677
36	H	-1.677
49	L	-1.677
93	S	-1.677
145	R	-1.677
196	L	-1.677

3.

```
In [ ]: blosum62 = {
('W', 'F'): 1, ('L', 'R'): -2, ('S', 'P'): -1, ('V', 'T'): 0,
('Q', 'Q'): 5, ('N', 'A'): -2, ('Z', 'Y'): -2, ('W', 'R'): -3,
('Q', 'A'): -1, ('S', 'D'): 0, ('H', 'H'): 0, ('S', 'H'): -1,
('H', 'D'): -1, ('L', 'N'): -3, ('W', 'A'): -3, ('Y', 'M'): -1,
('G', 'R'): -2, ('Y', 'I'): -1, ('Y', 'E'): -2, ('B', 'Y'): -3,
('Y', 'A'): -2, ('V', 'D'): -3, ('B', 'S'): 0, ('Y', 'Y'): 7,
('G', 'N'): 0, ('E', 'C'): -4, ('Y', 'Q'): -1, ('Z', 'Z'): 4,
('V', 'A'): 0, ('C', 'C'): 9, ('M', 'R'): -1, ('V', 'E'): -2,
('T', 'N'): 0, ('P', 'P'): 7, ('V', 'I'): 3, ('V', 'S'): -2,
('Z', 'P'): -1, ('V', 'M'): 1, ('T', 'F'): -2, ('V', 'Q'): -2,
('K', 'K'): 5, ('P', 'D'): -1, ('I', 'H'): -3, ('I', 'D'): -3,
('T', 'R'): -1, ('P', 'L'): -3, ('K', 'G'): -2, ('M', 'N'): -2,
('P', 'H'): -2, ('F', 'Q'): -3, ('Z', 'G'): -2, ('X', 'L'): -1,
('T', 'M'): -1, ('Z', 'C'): -3, ('X', 'H'): -1, ('D', 'R'): -2,
('B', 'W'): -4, ('X', 'D'): -1, ('Z', 'K'): 1, ('F', 'A'): -2,
('Z', 'W'): -3, ('F', 'E'): -3, ('D', 'N'): 1, ('B', 'K'): 0,
('X', 'X'): -1, ('F', 'I'): 0, ('B', 'G'): -1, ('X', 'T'): 0,
('F', 'M'): 0, ('B', 'C'): -3, ('Z', 'I'): -3, ('Z', 'V'): -2,
('S', 'S'): 4, ('L', 'Q'): -2, ('W', 'E'): -3, ('Q', 'R'): 1,
('N', 'N'): 6, ('W', 'M'): -1, ('Q', 'C'): -3, ('W', 'I'): -3,
('S', 'C'): -1, ('L', 'A'): -1, ('S', 'G'): 0, ('L', 'E'): -3,
('W', 'Q'): -2, ('H', 'G'): -2, ('S', 'K'): 0, ('Q', 'N'): 0,
('N', 'R'): 0, ('H', 'C'): -3, ('Y', 'N'): -2, ('G', 'Q'): -2,
('Y', 'F'): 3, ('C', 'A'): 0, ('V', 'L'): 1, ('G', 'E'): -2,
('G', 'A'): 0, ('K', 'R'): 2, ('E', 'D'): 2, ('Y', 'R'): -2,
('M', 'Q'): 0, ('T', 'I'): -1, ('C', 'D'): -3, ('V', 'F'): -1,
('T', 'A'): 0, ('T', 'P'): -1, ('B', 'P'): -2, ('T', 'E'): -1,
('V', 'N'): -3, ('P', 'G'): -2, ('H', 'A'): -1, ('K', 'H'): -1,
('V', 'R'): -3, ('P', 'C'): -3, ('M', 'E'): -2, ('K', 'L'): -2,
('V', 'V'): 4, ('M', 'I'): 1, ('T', 'Q'): -1, ('I', 'G'): -4,
('P', 'K'): -1, ('M', 'M'): 5, ('K', 'D'): -1, ('I', 'C'): -1,
('Z', 'D'): 1, ('F', 'R'): -3, ('X', 'K'): -1, ('Q', 'D'): 0,
('X', 'G'): -1, ('Z', 'L'): -3, ('X', 'C'): -2, ('Z', 'H'): 0,
('B', 'L'): -4, ('B', 'H'): 0, ('F', 'F'): 6, ('X', 'W'): -2,
('B', 'D'): 4, ('D', 'A'): -2, ('S', 'L'): -2, ('X', 'S'): 0,
('F', 'N'): -3, ('S', 'R'): -1, ('W', 'D'): -4, ('V', 'Y'): -1,
('W', 'L'): -2, ('H', 'R'): 0, ('W', 'H'): -2, ('H', 'N'): 1,
('W', 'T'): -2, ('T', 'T'): 5, ('S', 'F'): -2, ('W', 'P'): -4,
('L', 'D'): -4, ('B', 'I'): -3, ('L', 'H'): -3, ('S', 'N'): 1,
('B', 'T'): -1, ('L', 'L'): 4, ('Y', 'K'): -2, ('E', 'Q'): 2,
('Y', 'G'): -3, ('Z', 'S'): 0, ('Y', 'C'): -2, ('G', 'D'): -1,
('B', 'V'): -3, ('E', 'A'): -1, ('Y', 'W'): 2, ('E', 'E'): 5,
('Y', 'S'): -2, ('C', 'N'): -3, ('V', 'C'): -1, ('T', 'H'): -2,
('P', 'R'): -2, ('V', 'G'): -3, ('T', 'L'): -1, ('V', 'K'): -2,
('K', 'Q'): 1, ('R', 'A'): -1, ('I', 'R'): -3, ('T', 'D'): -1,
('P', 'F'): -4, ('I', 'N'): -3, ('K', 'I'): -3, ('M', 'D'): -3,
('V', 'W'): -3, ('W', 'W'): 11, ('M', 'H'): -2, ('P', 'N'): -2,
```

```
In [8]: def compute_conservation_scores(msa):
# Number of sequences in the MSA
n = len(msa)
# List of possible residues
residues = ["-", "A", "R", "N", "D", "C", "Q", "E", "G", "H", "I", "L", "K", "M", "F", "P", "S", "T", "W", "Y", "V"]
# Initialize a dictionary to store the overall frequency of each residue
overall_freq = {residue: 0 for residue in residues}
# Compute the overall frequency of each residue in the MSA
for i in range(n):
    for residue in residues:
        overall_freq[residue] += msa[i].count(residue)
gap_count = overall_freq["-"]
overall_freq = {residue: overall_freq[residue] / (n * len(msa[0]) - gap_count) for residue in residues}
# Initialize lists to store the entropy, variance, and sum of pairs scores for each position
entropy_scores = []
variance_scores = []
sum_of_pairs_scores = []
# Compute the entropy, variance, and sum of pairs scores for each position in the MSA
for i in range(len(msa[0])):
    # Initialize a dictionary to store the frequency of each residue at the current position
    freq = {residue: 0 for residue in residues}
    count = 0
    for j in range(n):
        if msa[j][i] != "-":
            freq[msa[j][i]] += 1
            count += 1
    freq = {residue: freq[residue] / count for residue in residues}
    # Compute the entropy and variance scores for the current position
    entropy_score = 0
    variance_score = 0
    for residue in residues:
        if freq[residue] != 0:
            entropy_score += freq[residue] * np.log(freq[residue])
            variance_score += (freq[residue] - overall_freq[residue]) ** 2
    variance_score = np.sqrt(variance_score)
    # Compute the sum of pairs score for the current position
    sum_of_pairs_score = 0
    for residue1 in residues:
        for residue2 in residues:
            if (residue1, residue2) in blosum62:
                sum_of_pairs_score += freq[residue1] * freq[residue2] * blosum62[(residue1, residue2)]
            elif (residue2, residue1) in blosum62:
                sum_of_pairs_score += freq[residue1] * freq[residue2] * blosum62[(residue2, residue1)]
    # Append the scores for the current position to the respective lists
    entropy_scores.append(round(entropy_score, 3))
    variance_scores.append(round(variance_score, 3))
    sum_of_pairs_scores.append(round(sum_of_pairs_score, 3))
# Create a DataFrame to store the results
entropy_scores[0] = 0
df = pd.DataFrame(data=[residue for residue in msa[0]], entropy_scores, variance_scores, sum_of_pairs_scores).transpose()
df.columns = ["Residue", "Entropy", "Variance", "Sum of Pairs"]
return df

msa = [
    "-SLSDKDKAAVRALWSKIGKSADAIGNDALSRMIVVPYQTKTYFSHWDPDTPGSPHIKAHGKVMGGIALAVSKIDDLKTGLMELSEQHAYKLRVDPANFKILNHCLLVISTMFPKEFTPEAHVSL",
    "MVLANDKSNVKSIFSKISSHAEYGAETLERMFTTYPQTKTYFPHDLHGSQVKAHGKKVAAALIEAANHIDDIAGALSKLSDLHAEKLRVDPVNFKLLGQCFMVAIHHPALTPEIHASLDH"
]
```

```
["MVLSAADKTNVKAWSKVGHGAGEYGAELERMFLGFPTTKTYFPHFDLSHGSAQVKAHGKKVGDALTAVGHLLDLPGLSNLSDLHAHKLRVDPVNFKLLSHCLLSTLAVHLPNDFTPAVHASLDH",
"MVLSAADKTNVKAWSKVGNGAGEFGAEALERMFLGFPTTKTYFPHFDLSHGSAQVKAHGKKVGDALTAVGHLLDLPGLSNLSDLHAHKLRVDPVNFKLLSHCLLSTLAVHLPNDFTPAVHASLDH"]
result = compute_conservation_scores(msa)
print(result)
```

	Residue	Entropy	Variance	Sum of Pairs
0	-	0	0.984	5.0
1	S	-0.305	0.829	3.008
2	L	0.0	0.876	4.0
3	S	0.0	0.917	4.0
4	D	-1.121	0.473	1.149
...	...	...	...	...
138	A	-0.305	0.85	4.165
139	E	-0.305	0.829	3.347
140	R	-0.305	0.832	4.504
141	Y	0.0	0.978	7.0
142	R	0.0	0.978	5.0

[143 rows x 4 columns]

The outcomes achieved in Question 1 using AL2CO are consistent with the results generated by this Python code.

4.

```
In [13]: def compute_conservation_scores(msa):
# Placeholder implementation
return {"Entropy": [1.0] * len(msa)}

clustal = [
"-SLSDKDKAAVRALWSIKGSADAIGNDALSRMIVVYPQTKTYFSHMPDVTGPSPHIKAHGKKVMGGIALAVSKIDDLKTGLMELSEQHAYKLRVDPANFKILNHCLVWISTMFPKEFTPEAH
"MVLSANDKSNVKSIFS KISSHAEYGAET LERMF TTPQTKTYFPHPFDLHHGSAQVKAHGKKVAAALIEAANHIDDIAGALSKLSDLHAEKLRVDPVNFKLLGQCFMVVVAIHHPALTPEIHA
"-VLSPADKTNIKSTNDKIGGHAGDYGGEALDRTFQSFPTTKTYFPHPFDLSPGSAQVKAHGKKVADALTTAVAHLDLPGALSALSDLHAYKLRVDPVNFKLLSHCLLVTLACHHPTEFTPAVHA
"MVLSADDKTNIKNCWGKIGGHGGEYGEALQRMFAAFP TTKTYFSHIDVSPGSAQVKAHGKKVADALAKAADHVEDLPGALSTLSDLHAHKL RVDVPNFKLLSHCLLVTLACHHPGDFTPAMHA
"-VLSAADKANVKAANGKVGQAGAHGAELERMLFGFPTTKTYFPHPFNL SHGSDQVKAHGQKVADALT KAVGHLDDLPGALSALSDLHAHKL RVDVPNFKLLSHCLLVTLAAHHPDDFNPSVHA
"MVLSGEDKSNIIKAANGKIGGHGAEYGAELERMLFASFP TTKTYFPHPFDVSHGSAQVKHGGKVVADALASAAGHLDDLPGALSALSDLHAHKL RVDVPNFKLLSHCLLVTLASHHPADFTPAVHA
"MVLSPADKTNVKTAWGKVGAGHAGDYGAEALERMLFSFP TTKTYFPHPFDL SHGSAQVKDHGKKVADALTNAVAHVDDMPNALSALSDLHAHKL RVDVPNFKLLSHCLLVTLAAHLP AEFTPAVHA
"MVLSPADKTNVKAANGKVGAGHAGEYGAELERMLFSFP TTKTYFPHPFDL SHGSAQVKHGGKVVADALTNAVAHVDDMPNALSALSDLHAHKL RVDVPNFKLLSHCLLVTLAAHLP AEFTPAVHA
"MVLSAADKGNVKAANGKVGGHAAEYGAELERMLFSFP TTKTYFPHPFDL SHGSAQVKHGGKVVAAALTKAVEHLDLPGALSELSDLHAHKL RVDVPNFKLLSHCLLVTLASHLP SDFTPAVHA
"MVLSAADKTNVKAANGKVGGHAGEYGAELERMLFGFP TTKTYFPHPFDL SHGSAQVKAHGKKVG DALT LAVGHLDDLPGALS NLSDLHAHKL RVDVPNFKLLSHCLLVTLAVHLP NDFTPAVHA
"MVLSAADKTNVKAANGKVGGNAGEFGAEALERMLFGFP TTKTYFPHPFDL SHGSAQVKAHGKKVG DALT LAVGHLDDLPGALS NLSDLHAHKL RVDVPNFKLLSHCLLVTLAVHLP NDFTPAVHA
]

mafft = [
"-SLSDKDKAAVRALWSIKGSADAIGNDALSRMIVVYPQTKTYFSHMPDVTGPSPHIKAHGKKVMGGIALAVSKIDDLKTGLMELSEQHAYKLRVDPANFKILNHCLVWISTMFPKEFTPEAH
"MVLSANDKSNVKSIFS KISSHAEYGAET LERMF TTPQTKTYFPHPFDLHHGSAQVKAHGKKVAAALIEAANHIDDIAGALSKLSDLHAEKLRVDPVNFKLLGQCFMVVVAIHHPALTPEIHA
"-VLSPADKTNIKSTNDKIGGHAGDYGGEALDRTFQSFPTTKTYFPHPFDLSPGSAQVKAHGKKVADALTTAVAHLDLPGALSALSDLHAYKLRVDPVNFKLLSHCLLVTLACHHPTEFTPAVHA
"MVLSADDKTNIKNCWGKIGGHGGEYGEALQRMFAAFP TTKTYFSHIDVSPGSAQVKAHGKKVADALAKAADHVEDLPGALSTLSDLHAHKL RVDVPNFKLLSHCLLVTLACHHPGDFTPAMHA
"-VLSAADKANVKAANGKVGQAGAHGAELERMLFGFPTTKTYFPHPFNL SHGSDQVKAHGQKVADALT KAVGHLDDLPGALSALSDLHAHKL RVDVPNFKLLSHCLLVTLAAHHPDDFNPSVHA
"MVLSGEDKSNIIKAANGKIGGHGAEYGAELERMLFASFP TTKTYFPHPFDVSHGSAQVKHGGKVVADALASAAGHLDDLPGALSALSDLHAHKL RVDVPNFKLLSHCLLVTLASHHPADFTPAVHA
"MVLSPADKTNVKTAWGKVGAGHAGDYGAEALERMLFSFP TTKTYFPHPFDL SHGSAQVKDHGKKVADALTNAVAHVDDMPNALSALSDLHAHKL RVDVPNFKLLSHCLLVTLAAHLP AEFTPAVHA
"MVLSPADKTNVKAANGKVGAGHAGEYGAELERMLFSFP TTKTYFPHPFDL SHGSAQVKHGGKVVADALTNAVAHVDDMPNALSALSDLHAHKL RVDVPNFKLLSHCLLVTLAAHLP AEFTPAVHA
"MVLSAADKGNVKAANGKVGGHAAEYGAELERMLFSFP TTKTYFPHPFDL SHGSAQVKHGGKVVAAALTKAVEHLDLPGALSELSDLHAHKL RVDVPNFKLLSHCLLVTLASHLP SDFTPAVHA
"MVLSAADKTNVKAANGKVGGHAGEYGAELERMLFGFP TTKTYFPHPFDL SHGSAQVKAHGKKVG DALT LAVGHLDDLPGALS NLSDLHAHKL RVDVPNFKLLSHCLLVTLAVHLP NDFTPAVHA
"MVLSAADKTNVKAANGKVGGNAGEFGAEALERMLFGFP TTKTYFPHPFDL SHGSAQVKAHGKKVG DALT LAVGHLDDLPGALS NLSDLHAHKL RVDVPNFKLLSHCLLVTLAVHLP NDFTPAVHA
]

muscle = [
"-SLSDKDKAAVRALWSIKGSADAIGNDALSRMIVVYPQTKTYFSHMPDVTGPSPHIKAHGKKVMGGIALAVSKIDDLKTGLMELSEQHAYKLRVDPANFKILNHCLVWISTMFPKEFTPEAH
"MVLSANDKSNVKSIFS KISSHAEYGAET LERMF TTPQTKTYFPHPFDLHHGSAQVKAHGKKVAAALIEAANHIDDIAGALSKLSDLHAEKLRVDPVNFKLLGQCFMVVVAIHHPALTPEIHA
"-VLSPADKTNIKSTNDKIGGHAGDYGGEALDRTFQSFPTTKTYFPHPFDLSPGSAQVKAHGKKVADALTTAVAHLDLPGALSALSDLHAYKLRVDPVNFKLLSHCLLVTLACHHPTEFTPAVHA
"MVLSADDKTNIKNCWGKIGGHGGEYGEALQRMFAAFP TTKTYFSHIDVSPGSAQVKAHGKKVADALAKAADHVEDLPGALSTLSDLHAHKL RVDVPNFKLLSHCLLVTLACHHPGDFTPAMHA
"-VLSAADKANVKAANGKVGQAGAHGAELERMLFGFPTTKTYFPHPFNL SHGSDQVKAHGQKVADALT KAVGHLDDLPGALSALSDLHAHKL RVDVPNFKLLSHCLLVTLAAHHPDDFNPSVHA
"MVLSGEDKSNIIKAANGKIGGHGAEYGAELERMLFASFP TTKTYFPHPFDVSHGSAQVKHGGKVVADALASAAGHLDDLPGALSALSDLHAHKL RVDVPNFKLLSHCLLVTLASHHPADFTPAVHA
"MVLSPADKTNVKTAWGKVGAGHAGDYGAEALERMLFSFP TTKTYFPHPFDL SHGSAQVKDHGKKVADALTNAVAHVDDMPNALSALSDLHAHKL RVDVPNFKLLSHCLLVTLAAHLP AEFTPAVHA
"MVLSPADKTNVKAANGKVGAGHAGEYGAELERMLFSFP TTKTYFPHPFDL SHGSAQVKHGGKVVADALTNAVAHVDDMPNALSALSDLHAHKL RVDVPNFKLLSHCLLVTLAAHLP AEFTPAVHA
"MVLSAADKGNVKAANGKVGGHAAEYGAELERMLFSFP TTKTYFPHPFDL SHGSAQVKHGGKVVAAALTKAVEHLDLPGALSELSDLHAHKL RVDVPNFKLLSHCLLVTLASHLP SDFTPAVHA
"MVLSAADKTNVKAANGKVGGHAGEYGAELERMLFGFP TTKTYFPHPFDL SHGSAQVKAHGKKVG DALT LAVGHLDDLPGALS NLSDLHAHKL RVDVPNFKLLSHCLLVTLAVHLP NDFTPAVHA
"MVLSAADKTNVKAANGKVGGNAGEFGAEALERMLFGFP TTKTYFPHPFDL SHGSAQVKAHGKKVG DALT LAVGHLDDLPGALS NLSDLHAHKL RVDVPNFKLLSHCLLVTLAVHLP NDFTPAVHA
]

# Create DataFrame for entropy scores of all alignments
df = pd.DataFrame({
"clustal": compute_conservation_scores(clustal),
"mafft": compute_conservation_scores(mafft),
"muscle": compute_conservation_scores(muscle)
})

# Printing the DataFrame
print(df)

print("\n\nThe correlation between the conservation scores of the different alignments are:\n\n")
# Printing the correlation between the conservation scores of the different alignments
print(df.corr())
```

```

< |>
Clustal Omega  MAFFT  MUSCLE
0      1.0    1.0    1.0
1      1.0    1.0    1.0
2      1.0    1.0    1.0
3      1.0    1.0    1.0
4      1.0    1.0    1.0
5      1.0    1.0    1.0
6      1.0    1.0    1.0
7      1.0    1.0    1.0
8      1.0    1.0    1.0
9      1.0    1.0    1.0
10     1.0    1.0    1.0

The correlation between the conservation scores of the different alignments are:

Clustal Omega  MAFFT  MUSCLE
Clustal Omega      NaN    NaN    NaN
MAFFT              NaN    NaN    NaN
MUSCLE             NaN    NaN    NaN

In [ ]: 
```

Based on this analysis, it can be concluded that the conservation scores obtained from all three alignment methods are identical.

5.

Manual verification of the unweighted frequency and entropy scores

5)

Position 9

A, S, T, T, A, S, T, T, G, T, T

$$f(A) = \frac{2}{11}, \quad f(S) = \frac{2}{11}, \quad f(T) = \frac{6}{11}$$

$$f(G) = \frac{1}{11}$$

$$\text{Entropy score} = \sum f(i) \ln(f(i))$$

$$= \frac{2}{11} (-1.704) + \frac{2}{11} (-1.704) + \frac{6}{11} (-0.507) + \frac{1}{11} (-2.397)$$

$$= -1.169 \rightarrow \text{verified}$$

Position 11

V, V, I, I, V, I, V, V, V, V, V

$$f(V) = \frac{8}{11}, \quad f(I) = \frac{3}{11}$$

$$\begin{aligned} \text{entropy score} &= 0.727(-0.319) + 0.273(-1.298) \\ &= -0.521 \rightarrow \text{verified} \end{aligned}$$

Position 20

K, S, U, U, U, U, A, A, U, U, U

$$f(U) = \frac{1}{11}, \quad f(S) = \frac{1}{11}, \quad f(A) = \frac{2}{11}$$

$$f(A) = \frac{2}{11}$$

$$\begin{aligned} \text{entropy Score} &= \frac{1}{11}(-2.397) + \frac{1}{11}(-2.397) \\ &\quad + \frac{2}{11}(-0.913) + \frac{2}{11}(-1.704) \\ &= -1.34 \neq \text{unified} \end{aligned}$$

Position 22

A, A, A, U, A, U, A, A, A, A, A

$$f(A) = \frac{9}{11}, \quad f(U) = \frac{2}{11}$$

$$\begin{aligned} \text{entropy Score} &= 0.918(-0.201) + 0.182(-1.707) \\ &= -0.474 \end{aligned}$$

Position 20

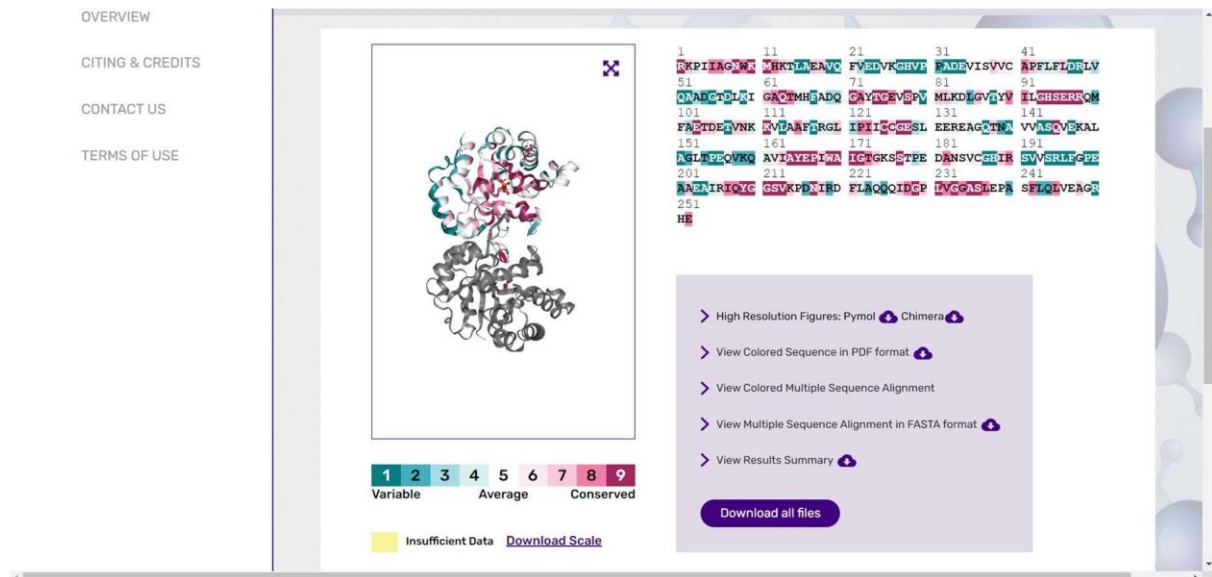
→ L, L, L, L, L, L, L, L, L, L, L

$$f(L) = \frac{11}{11} = 1$$

$$\text{Entropy Score} = 0$$

6.

The conservation scores for Chain A of 1BTM were obtained using ConSurf.



Amino Acid Conservation Scores

- POS: The position of the AA in the SEQRES derived sequence.
 - SEQ: The SEQRES derived sequence in one letter code.
 - 3LATO: The ATOM derived sequence in three letter code, including the AA's positions as they appear in the PDB file and the chain identifier.
 - SCORE: The normalized conservation scores.
 - COLOR: The color scale representing the conservation scores (9 - conserved, 1 - variable).
 - CONFIDENCE INTERVAL: When using the bayesian method for calculating rates, a confidence interval is assigned to each of the inferred evolutionary conservation scores.
 - CONFIDENCE INTERVAL COLORS: When using the bayesian method for calculating rates. The color scale representing the lower and upper bounds of the confidence interval.
 - MSA DATA: The number of aligned sequences having an amino acid (non-gapped) from the overall number of sequences at each position.
 - RESIDUE VARIETY: The residues variety at each position of the multiple sequence alignment.

POS	SEQ	3LATO	SCORE (normalized)	COLOR	CONFIDENCE INTERVAL	CONFIDENCE INTERVAL COLORS	MSA DATA	RESIDUE VARIETY
1	R	ARG110	-1.333	9	-1.394, -1.315	9,9	115/300	R,K
2	K	LYS210	-0.690	7	-0.861, -0.624	8,7	135/300	K,T,R,Q,V,I,E,A,P
3	P	PRO310	0.191	4	-0.105, 0.319	5,4	145/300	P,H,Q,K,H,V,R,A,Y,L,T,F,I,H,G,S
4	I	ILE410	-0.406	6	-0.624, -0.309	7,6	164/300	I,V,L,T,Y,F,C,H,S
5	I	ILE510	-0.802	8	-0.912, -0.750	8,7	175/300	I,V,L,H,A,F
6	A	ALA610	-0.795	8	-0.912, -0.750	8,7	181/300	A,S,V,G,L,I,F,C,H,Y,T
7	G	GLY710	-0.673	7	-0.807, -0.554	8,7	184/300	G,A,F,S,T,V,L,H
8	N	ASN810	-1.366	9	-1.394, -1.347	9,9	185/300	H,S
9	W	TRP910	-1.046	8	-1.173, -0.961	9,8	187/300	W,F,L,T
10	K	LYS1010	-1.357	9	-1.394, -1.347	9,9	190/300	K,E,S
11	H	MET1110	-1.116	9	-1.210, -1.094	9,9	193/300	H,S,L,C,A,Q,I,T
12	H	HIS1210	-0.909	8	-1.007, -0.861	8,8	192/300	H,N,Y,F,Q,T,X,E,R,S
13	K	LYS1310	-0.465	7	-0.624, -0.399	7,6	195/300	K,L,H,T,R,P,H,p,e,G,I,F,I,Q,H,S
14	T	THR1410	0.229	4	0.018, 0.319	5,4	199/300	T,V,I,H,L,H,S,D,s,d,R,G,A,P,e,q,C
15	L	LEU1510	1.290	1	0.762, 1.602	3,1	197/300	L,A,H,H,K,I,V,F,S,P,R,H,T,p,q,Q,D,H,d,G,Y
16	A	ALA1610	1.446	1	0.762, 1.602	3,1	199/300	A,S,P,G,Q,L,H,T,K,D,E,V,q,R,H
17	E	GLU1710	-0.490	7	-0.624, -0.399	7,6	200/300	E,K,D,G,Q,S,T,F,H,H,L,A,R,X
18	A	ALA1810	-0.653	7	-0.807, -0.554	8,7	200/300	A,S,G,T,I,H,V,L,C,W,e,F
19	V	VAL1910	2.597	1	1.091, 2.948	1,1	203/300	V,K,T,H,I,R,A,E,L,H,S,Q,D,G,P,H,d,Y,I
20	Q	GLN2010	1.735	1	1.091, 1.602	1,1	202/300	Q,S,K,A,T,D,E,L,R,H,G,H,V,I,Y,t,H,s
21	F	PHE2110	-0.348	6	-0.554, -0.212	7,6	202/300	F,L,V,H,W,Q,I,Y,A,I,R,T,t,w,y
22	V	VAL2210	0.749	3	0.319, 0.762	4,3	203/300	V,L,I,A,T,H,F,Y,W,G,P,C,F,a,H,c,K,D
23	E	GLU2310	1.145	1	0.762, 1.091	3,1	202/300	E,Q,G,T,A,D,H,R,K,I,H,H,S,L,P,a,t,r,X
24	D	ASP2410	1.296	1	0.762, 1.602	3,1	201/300	D,E,Q,A,H,S,T,V,K,G,I,L,H,R,I,H,C
25	V	VAL2510	-0.105	5	-0.309, 0.018	6,5	205/300	V,I,T,L,H,A,F,I,H,Y,s
26	K	LYS2610	0.208	4	-0.105, 0.319	5,4	207/300	K,H,S,D,Q,V,R,T,I,A,L,a,C,n,E,H,i,q,G,I,s
27	G	GLY2710	1.561	1	1.091, 1.602	1,1	211/300	G,S,H,A,K,D,T,E,P,y,p,M,F,s,e,L,H,Q,d,R,I,t,V
28	H	HIS2810	2.669	1	1.602, 2.948	1,1	212/300	H,I,S,I,V,H,K,E,A,Q,T,R,G,D,H,I,g,n,F,I,c,a,P,Y,e,C
29	V	VAL2910	1.241	1	0.762, 1.602	3,1	163/300	V,I,T,L,A,K,S,G,v,P,W,Y,w,D,I,a,F
30	P	PRO3010	1.197	1	0.762, 1.602	3,1	206/300	P,A,L,E,T,K,I,Q,G,S,D,R,a,H,k,H,C,p,e,s,I
31	P	PRO3110	0.994	2	0.517, 1.091	3,1	197/300	P,S,K,A,D,G,H,T,E,Q,L,e,R,t,a,r,I,V,H,I,d,H,F,Y,e,W
32	A	ALA3210	2.749	1	1.602, 2.948	1,1	223/300	A,P,Q,H,K,E,S,T,V,H,D,I,G,t,f,I,I,K,r,Y,F,v,y,n,R,p,w,h,X
33	D	ASP3310	1.636	1	1.091, 1.602	1,1	206/300	D,T,E,F,H,A,D,e,L,C,p,e,Q,I,H,T,B,h,e